

LECTURE 9: CREATING DATA TYPES

- Reading:** Supplements for Lecture 9
10 min
- Video:** Overview
5 min
- Video:** Point charges
20 min
- Video:** Turtle graphics
10 min
- Video:** Complex numbers
22 min
- Reading:** Optional Enrichment on Creating Data Types
10 min
- Quiz:** Creating Data Types
Submitted
- Programming Assignment:** Creating Data Types
8h

Programming Assignment: Creating Data Types

✔ Passed • 91/100 points

Deadline Pass this assignment by Nov 22, 11:59 PM IST

Instructions	My submissions	Discussions
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Create submission +

Date	Score	Passed
November 3, 2023 6:21 PM IST	91/100	Yes

Creating Data Types	91/100	Hide grader output
See the Assessment Guide for information on how to interpret this report. ASSESSMENT SUMMARY Compilation: PASSED API: PASSED SpotBugs: FAILED (2 warnings) PMD: FAILED (3 warnings) Checkstyle: FAILED (0 errors, 24 warnings) Correctness: 29/34 tests passed Memory: 2/2 tests passed Timing: 51/51 tests passed		
Aggregate score: 91.18%		

November 3, 2023 5:58 PM IST	88/100	Yes
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Creating Data Types	88/100	Show grader output
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November 3, 2023 5:39 PM IST	10/100	No
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Creating Data Types	10/100	Show grader output
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November 3, 2023 5:34 PM IST	5/100	No
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Creating Data Types	5/100	Show grader output
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👍 Like👎 Dislike📄 Report an issue

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Checking the APIs of your programs.
*-----

ColorHSB:

Clock:

=====

*****
*   CHECKING STYLE AND COMMON BUG PATTERNS
*****

% spotbugs *.class
*-----
M D DLS_DEAD_LOCAL_STORE_SHADOWS_FIELD DLS: Assigns a value to the local variable 'm', but that value is never used. Did you mean to assign the v
M D DLS_DEAD_LOCAL_STORE_SHADOWS_FIELD DLS: Assigns a value to the local variable 'h', but that value is never used. Did you mean to assign the v
SpotBugs ends with 2 warnings.

=====

% pmd .
*-----
ColorHSB.java:8: The private instance (or static) variable 'b' can be made 'final'; it is initialized only in the declaration or constructor. [Im
ColorHSB.java:8: The private instance (or static) variable 'h' can be made 'final'; it is initialized only in the declaration or constructor. [Im
ColorHSB.java:8: The private instance (or static) variable 's' can be made 'final'; it is initialized only in the declaration or constructor. [Im
PMD ends with 3 warnings.

=====

% checkstyle *.java
*-----
[WARN] Clock.java:10:33: The local (or parameter) variable 'h' has the same name as an instance variable. Use a different name. [HiddenField]
[WARN] Clock.java:10:40: The local (or parameter) variable 'm' has the same name as an instance variable. Use a different name. [HiddenField]
[WARN] Clock.java:16:5: Define constructors after static and instance variables but before methods. [DeclarationOrder]
[WARN] Clock.java:27:5: Define constructors after static and instance variables but before methods. [DeclarationOrder]
[WARN] Clock.java:28:11: The local (or parameter) variable 'h' has the same name as an instance variable. Use a different name. [HiddenField]
[WARN] Clock.java:28:14: The local (or parameter) variable 'm' has the same name as an instance variable. Use a different name. [HiddenField]
[WARN] Clock.java:29:13: Inner assignments should be avoided. [InnerAssignment]
[WARN] Clock.java:37:26: '!=' is not followed by whitespace. [WhitespaceAround]
Checkstyle ends with 0 errors and 8 warnings.

% custom checkstyle checks for ColorHSB.java
*-----
[WARN] ColorHSB.java:6: The 'main()' method must call either 'StdIn.isEmpty()' or 'StdIn.hasNextLine()' to determine when it has finished reading. [MainMethodCallCount]
[WARN] ColorHSB.java:6: The number (6) of calls to 'Integer.parseInt()' must equal the number (3) of integer command-line arguments. [CommandLineArgumentCount]
[WARN] ColorHSB.java:6: You must call 'StdIn.readString()' at least once from 'main()'. [MethodCallCount]
[WARN] ColorHSB.java:6: You should call 'StdIn.readString()' at least 2 times from 'main()'. [MethodCallCount]
```